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Assignment of bachelor's thesis

Title:	Transformer-Based Classification of Astronomical Light Curves
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Study program:	Informatics
Branch / specialization:	Artificial Intelligence 2021
Department:	Department of Applied Mathematics
Validity:	until the end of summer semester 2026/2027

Instructions

Introduction

Transformer architectures employing the self-attention mechanism represent one of the most effective and best-performing machine learning approaches available today. Their ability to model dependencies across both local and distant segments of an input sequence makes them particularly well suited to classifying a wide range of stellar variability in extensive surveys of light curves (time series of changes in stellar brightness). Despite several published studies, the use of transformer-based architectures for light-curve classification is still limited, and numerous methodological issues remain unresolved.

Thesis Objective

The primary objective of this bachelor's thesis is to design, implement, and evaluate a pretrained transformer-based architecture for supervised classification of variable stars based on their high-precision light curves obtained with TESS satellite. Particular emphasis will be placed on discriminating between distinct classes of eclipsing, pulsating, and rotating stars using their characteristic time-domain variability patterns.

Proposed Tasks

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1) Examine existing pretrained transformer architectures, such as Vision Transformers adapted for one-dimensional sequences or large language models with tailored tokenizers, and identify the most appropriate approach for analyzing stellar light curves as time series.

2) Gain practical experience with the Hugging Face ecosystem, including model and dataset hubs, create a free account and select a suitable pretrained model and time-series dataset that closely match the intended application.

3) Identify in the literature an appropriate sample of TESS light curves for which a manual classification has already been performed.

4) Set up an experimental environment with GPU support. For smaller experiments, use the servers at the Astronomical Institute of the Czech Academy of Sciences; for larger experiments, use your account on the RCI cluster.

5) Design an effective preprocessing pipeline for TESS light curves, including standardization of the time sampling, flux detrending and normalization, and conversion into a format suitable as input to the transformer-based model.

6) Perform multiple machine learning experiments using diverse model configurations and hyperparameters, and evaluate their performance with relevant quantitative metrics for multi-class classification.

7) Compare the results with a simple baseline model as well as with object classes published in astronomical catalogues.

8) Evaluate the suitability of the proposed transformer-based framework for large-scale classification of variable stars in terms of both predictive performance and computational cost, identify bottlenecks, and suggest directions for future improvements.

The recommended literature will be provided by the supervisor.

Electronically approved by Ing. Magda Friedjungová, Ph.D. on 19 December 2025 in Prague.